

Air Conditioners Data Analysis

Stage 5: Data Aggregation & Representation

This notebook is independently executable in Google Colab. Every step explains what we do, why we do it, what we obtain, and how the result is used in the next stage.

Stage Objective

Stage 5 performs grouping, counting and aggregation so that large numbers of individual product records can be summarized meaningfully.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the project CSV directly so this notebook is independently executable.
csv_path = "/content/Air_Conditioners_Cleaned.csv"
df = pd.read_csv(csv_path)

print("CSV loaded successfully.")
print("Dataset shape:", df.shape)
```

```
CSV loaded successfully.
Dataset shape: (720, 14)
```

Why Aggregation Is Needed

Individual rows answer product-level questions. Aggregation answers questions about groups, such as how many products belong to a category or what the average numeric value is for a group.

```
cats=df.select_dtypes(include=['object','category','bool']).columns.tolist(); print(cats)
```

```
['name', 'main_category', 'sub_category', 'image', 'link', 'brand']
```

Reload the CSV

The dataset is loaded independently and the fields required for aggregation are identified.

```
nums=df.select_dtypes(include=np.number).columns.tolist(); print(nums)
```

```
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
```

Choose Grouping Fields

Categorical fields are examined and an actual field from the dataset is selected for group-level summaries.

```
if cats: counts=df.groupby(cats[0],dropna=False).size().reset_index(name='count'); print(counts)
```

	name	count
0	1 Ton 3 Star Split System Air Conditioner With...	1
1	1 Ton 3 Star Twin Inverter Split Fastcool Silv...	1
2	1 Ton 3 Star Window AC (Copper, WFA312LL, 2022...	1
3	1.5 Ton 2 Star Inverter Split System Air Condi...	1
4	1.5 Ton 3 Star Inverter Split AC (Copper) 2.5 ...	1
..	...	...
703	Ximoon Mini Air Cooler Humidifier Leakproof & ...	1
704	YONCDIAT Upgraded Personal Air Conditioner Co...	1
705	ZAAHN Ultra Chill Personal Cooler and Humidifi...	1
706	iMee-1.5 Ton Inverter Split Air Conditioner wi...	1
707	smartmi Air Purifiers for Home, Works with Hom...	1

```
[708 rows x 2 columns]
```

Count by Category

The number of records in each available category is calculated.

```
if cats and nums: summary=df.groupby(cats[0],dropna=False)[nums[0]].agg(['count','mean','min','max','sum']).reset_index(); print(summary)
```

	name	count	mean	min	max	\
0	1 Ton 3 Star Split System Air Conditioner With...	1	4.0	4.0	4.0	
1	1 Ton 3 Star Twin Inverter Split Fastcool Silv...	1	4.0	4.0	4.0	
2	1 Ton 3 Star Window AC (Copper, WFA312LL, 2022...	1	5.0	5.0	5.0	
3	1.5 Ton 2 Star Inverter Split System Air Condi...	1	4.0	4.0	4.0	
4	1.5 Ton 3 Star Inverter Split AC (Copper) 2.5 ...	1	4.0	4.0	4.0	
..	...	...	...	...	...	
703	Ximoon Mini Air Cooler Humidifier Leakproof & ...	1	4.0	4.0	4.0	
704	YONCDIAT Upgraded Personal Air Conditioner Co...	1	3.2	3.2	3.2	
705	ZAAHN Ultra Chill Personal Cooler and Humidifi...	1	3.3	3.3	3.3	
706	iMee-1.5 Ton Inverter Split Air Conditioner wi...	1	4.0	4.0	4.0	
707	smartmi Air Purifiers for Home, Works with Hom...	1	4.0	4.0	4.0	

	sum
0	4.0
1	4.0
2	5.0
3	4.0
4	4.0
..	...

```
703 4.0
704 3.2
705 3.3
706 4.0
707 4.0

[708 rows x 6 columns]
```

Calculate Numeric Summaries

For a suitable numeric field, count, mean, minimum, maximum and sum are calculated by group.

```
if 'summary' in locals(): print(summary.sort_values('mean',ascending=False))

      name  count  mean  min  max  \
34  AC Stand/Heavy Duty Air Conditioner Outdoor Un...      1  5.0  5.0  5.0
683 Whirlpool 1.5 Ton 3 Star Inverter Split AC (3D...      1  5.0  5.0  5.0
470 Panasonic 1.5 Ton 3 Star Hot and Cold Wi-Fi In...      1  5.0  5.0  5.0
494 Panasonic 2.2 Ton Inverter 3 Star Split Smart ...      1  5.0  5.0  5.0
468 Panasonic 1.0 Ton Inverter 3 Star Copper 2022 ...      1  5.0  5.0  5.0
..      ...      ...      ...      ...
580 TOSHIBA 1.5 Ton 3 Star Inverter Split AC (Copp...      1  1.0  1.0  1.0
52  Belity Portable Mini Air Conditioner Desktop A...      1  1.0  1.0  1.0
92  CARRIER SPLIT AC 1 TON 3 STAR FIX SPEED (12K 3...      1  1.0  1.0  1.0
133 Carrier Emperia Cxi 18K 1.5 Ton 3 Star Hybridj...      1  1.0  1.0  1.0
153 Daikin 1 Ton 4 Star Inverter Split AC (Copper,...      1  1.0  1.0  1.0

      sum
34  5.0
683 5.0
470 5.0
494 5.0
468 5.0
..  ...
580 1.0
52  1.0
92  1.0
133 1.0
153 1.0

[708 rows x 6 columns]
```

Sort the Summary

The aggregated table is sorted so differences between groups are easier to inspect.

```
if cats: print(df[cats[0]].value_counts(dropna=False).head(15))

name
Samsung 1.5 Ton 3 Star Wind-Free Technology Inverter Split AC (Copper, Convertible 5-in-1 Cooling Mode, Anti-bacterial Fil...      3
White Westinghouse 1.5 ton 3 Star fix Speed Split AC      3
LG 1.5 Ton 5 Star AI DUAL Inverter Split AC (Copper, Super Convertible 6-in-1 Cooling, HD Filter with Anti-Virus Protectio...      2
VESTAR 1.0Ton 3 Star Split AC (100% Copper Coil with Hidden Display Eco/Power Saving Mode, White)      2
Samsung 1.5 Ton 3 Star Wi-Fi Enabled, Wind-Free Technology Inverter Split AC (Copper, Convertible 5-in-1 Cooling Mode, Ant...      2
LG 2.0 Ton 3 Star AI DUAL Inverter Split AC (Copper, Super Convertible 6-in-1 Cooling, 4 Way Swing, HD Filter with Anti-Vi...      2
Panasonic 1 Ton 5 Star Wi-Fi Inverter Smart Split AC (Copper, 7 in 1 Convertible with additional AI Mode, 4 Way Swing, nan...      2
Personal Air Conditioners Portable AC Mini Air Conditioner for Office Dorm Room,Small AC Desktop Air Cooler Fan with Humid...      2
Panasonic 1.5 Ton 5 Star Wi-Fi Inverter Smart Split AC (Copper, 7 in 1 Convertible with additional AI Mode, 4 Way Swing, n...      2
Electrolux 1.5 Ton 5 Star Convertible Inverter Split Air Conditioner (AC), 100% Copper, 4 Step Filtration with PM2.5 Filte...      2
Master Electronic provides 1.5 Ton 3 Star Dust Filter Window AC (Copper, 2020 FRWL50TV162, White)      1
AC outdoor stand for 1 ton and 1.5 ton      1
Panasonic 2 Ton 4 Star Wi-Fi Twin-Cool Inverter Split Air Conditioner (Copper, 7 In 1 Convertible with additional AI Mode,...      1
Panasonic 1.5 Ton 4 Star Wi-Fi Twin-Cool Inverter Split Air Conditioner (Copper, 7 In 1 Convertible with additional AI Mod...      1
LG PS-Q19ENYE 1.5 Ton 4 Star Split AC      1
Name: count, dtype: int64
```

Compare Group Sizes

Category counts show where the dataset has more or fewer observations. This helps contextualize later averages.

```
if 'summary' in locals(): print(summary.describe(include='all').T)

      count  unique      top freq  \
name      708      708 smartmi Air Purifiers for Home, Works with Hom...      1
count      708.0      NaN      NaN      NaN
mean      708.0      NaN      NaN      NaN
min      708.0      NaN      NaN      NaN
max      708.0      NaN      NaN      NaN
sum      708.0      NaN      NaN      NaN

      mean      std  min  25%  50%  75%  max
name      NaN      NaN  NaN  NaN  NaN  NaN  NaN
count  1.016949  0.149476  1.0  1.0  1.0  1.0  3.0
mean  3.886088  0.614621  1.0  3.9  4.0  4.0  5.0
min    3.88404  0.615833  1.0  3.9  4.0  4.0  5.0
max    3.888136  0.615018  1.0  3.9  4.0  4.0  5.0
sum    3.954096  0.870268  1.0  3.9  4.0  4.0  12.0
```

Create Reusable Summary Tables

Compact summary DataFrames are created so Stage 6 can analyse them and Stage 7 can visualise them.

```
if 'summary' in locals(): print('Summary rows:',len(summary))

Summary rows: 708
```

Validate Aggregation

The summary table is inspected for shape and values so grouping errors are easier to detect.

```
print('Aggregation stage completed.')
```

```
Aggregation stage completed.
```